# A Gini-Frontier Compact Certification Framework for Exact Static Rebalancing

Algorithm chapter generated from the audited stable implementation

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**Abstract**

We present the stable paper-facing exact method implemented by `paper-gf-tailored-bc`. The method combines a Gini-frontier decomposition, valid interval screening, and CPLEX-managed fixed-interval compact branch-and-cut. A relaxation vector selects bounded route-cutset candidates, while validity of every added inequality is independent of that vector. A full-frontier ledger, independent incumbent verification, and source audits determine whether a global claim is admissible.

## 1 Problem Definition

Let $\mathcal{I} = \{1, \ldots, n\}$ be the stations, let $\mathcal{K}$ be the vehicles, and let 0 denote the depot. Station $i$ has initial inventory $b_i$, capacity $c_i$, positive target $t_i$, and penalty weight $\omega_i$. Vehicle $k$ has capacity $Q_k$ and duration limit $T_k$. Travel times are $\tau_{ij}$.

The compact model uses binary arc variables $x_{kij}$ and visit variables $z_{ki}$, nonnegative integer pickup and drop quantities $p_{ki}$ and $d_{ki}$, and final station inventory

$$Y_i = b_i + \sum_{k \in \mathcal{K}} (d_{ki} - p_{ki}), \qquad 0 \le Y_i \le c_i. \tag{1}$$

Standard depot flow, station flow, visit–service linking, station disjointness, vehicle load conservation, and load-capacity constraints make the selected arcs and operations an original feasible route-load plan. Vehicles start empty and may return residual load to the depot. Under the implemented handling convention, route duration is

$$\sum_{i,j} \tau_{ij} x_{kij} + c^{\text{unit}} \sum_i p_{ki} \le T_k, \qquad c^{\text{unit}} = c^{\text{pick}} + c^{\text{drop}}. \tag{2}$$

Thus a global handling inequality may charge pickups only; charging both $p$ and $d$ by $c^{\text{unit}}$ would not be valid for this convention.

Define the target ratios, denominator, pairwise deviations, and numerator by

$$r_i = Y_i / t_i, \quad S = \sum_i r_i, \quad h_{ij} = |r_i - r_j| \ (i < j), \quad H = \sum_{i<j} h_{ij}. \tag{3}$$

Because $t_i > 0$ and the feasible model maintains positive aggregate station inventory, $S > 0$. The Gini term [1] and weighted target-deviation penalty are

$$G = \frac{H}{nS}, \qquad P = \sum_i \omega_i e_i, qquad e_i = |r_i - 1|. \tag{4}$$

For fixed $\lambda \geq 0$, the objective is

$$F = G + \lambda P. \tag{5}$$

The binary-expansion compact formulation represents the products required by the Gini objective to the declared numerical tolerance. An independently reconstructed route-load plan is evaluated directly with (4)–(5); only such a verified plan can supply an upper bound.

**Definition 1** (Original feasible set)**.** $\mathcal{X}$ is the set of route, service, load, and final-inventory vectors that satisfy the original compact routing and operation constraints and the duration convention (2). Strengthening rows do not redefine $\mathcal{X}$; each paper-core row below is necessary for membership in $\mathcal{X}$ under its stated interval condition.

## 2 Gini-Frontier Certification Framework

Suppose a verifier-gated incumbent has objective $\overline{F}$. Any improving solution satisfies $G < \overline{F}$ because $P \geq 0$. The algorithm partitions the relevant improving Gini domain into closed intervals $\Gamma_q = [\gamma_q^L, \gamma_q^U]$ whose interiors are disjoint and whose union covers that domain.

For each interval, a valid non-enumerative relaxation first supplies a lower bound. An interval not fathomed by that bound is passed to the original fixed-interval compact model. Its no-improver form includes

$$\gamma_q^L \leq G \leq \gamma_q^U, \tag{6}$$

$$G + \lambda P \leq \overline{F} - \varepsilon. \tag{7}$$

CPLEX then proves infeasibility, proves a valid best bound at least $\overline{F} - \varepsilon$, or returns an unresolved leaf with its valid bound. A timeout or wrapper checkpoint never becomes an exact closure by itself.

**Proposition 2** (Fixed-interval no-improver validity)**.** *If the model consisting of the original compact formulation,* (6), (7)*, and globally valid strengthening rows is infeasible, then no original feasible solution in* $\Gamma_q$ *improves the incumbent.*

*Proof.* Every original improving solution in $\Gamma_q$ satisfies the original model, the interval rows, and (7). Every strengthening row is a necessary condition for that same solution. Such a solution would therefore be feasible in the fixed-interval model, contradicting infeasibility. □

**Theorem 3** (Full-frontier global exactness)**.** *Let* $\widehat{x} \in \mathcal{X}$ *be independently verified with objective* $\overline{F}$*. Assume (i) the final Gini leaves exactly cover every possible improving Gini value; (ii) each final leaf is closed by a valid relaxation bound, exact fixed-interval infeasibility, a valid fixed-interval best bound, or an empty-domain proof; and (iii) no invalid, unresolved, or open leaf remains. Then* $\widehat{x}$ *is globally optimal.*

*Proof.* If a better solution existed, its Gini value would belong to one final leaf. The corresponding leaf closure proves that no improving original solution exists in that leaf, a contradiction. Hence no feasible solution has objective below $\overline{F}$. □

An optional paper-safe $S$ refinement may partition a fixed Gini leaf's valid domain $[S^L, S^U]$ into children. This option was disabled in the tested route profile.

**Proposition 4** (Optional $S$-ledger disjunction)**.** *If* $[S^L, S^U] = \bigcup_b [S_b^L, S_b^U]$ *with disjoint interiors and every child is exactly closed using the enforced child bounds, then the parent Gini leaf is closed.*

*Proof.* Every parent-feasible point has one denominator value $S$ and therefore lies in at least one child. Closure of every child excludes every parent-feasible improver. Exact coverage prevents an omitted denominator region. $\square$

> **Stable algorithm.** Generate and verify a same-run heuristic incumbent; construct the full Gini ledger; screen leaves with valid relaxation bounds; solve unresolved leaves with the tailored compact model and the selected route-cutset callback; merge only exact leaf evidence; certify only after the coverage and source audits pass.

Operationally, the stable implementation executes the following generic sequence for every instance:

1. construct and independently verify a same-run HGA–TGBC incumbent;
2. partition the complete improving Gini domain and initialize its ledger;
3. apply valid non-enumerative bounds and domain propagation to each leaf;
4. solve every unresolved leaf with the original fixed-interval compact model, paper-safe static rows, and the bounded route-cutset separator;
5. merge only scope-valid leaf bounds or exact closures; and
6. issue a global certificate only after coverage, verifier, source, and finalization audits all pass.

# 3 Active Paper-Safe Strengthening

This section states the families enabled by the tested `structural_route_limited` profile. A family may add zero rows on a particular leaf; activation means that its separator or model generator is available under the conditions below.

## 3.1 Gini and product rows

**Proposition 5** (Direct interval rows)**.** *For a fixed interval, the inequalities*

$$H \le n\gamma^U S, \qquad H \ge n\gamma^L S \tag{8}$$

*are valid.*

*Proof.* Multiply $\gamma^L \le H/(nS) \le \gamma^U$ by the positive value $nS$. $\square$

**Proposition 6** (Interval-tight binary products)**.** *Let $a \in \{0,1\}$, $G \in [\gamma^L, \gamma^U]$, and $w = Ga$. The four rows*

$$w \ge \gamma^L a, \qquad w \le \gamma^U a, \tag{9}$$
$$w \ge G - \gamma^U(1-a), \qquad w \le G - \gamma^L(1-a) \tag{10}$$

*describe $w = Ga$ exactly for binary $a$ and strengthen a generic $[0,1]$ envelope.*

*Proof.* For $a = 0$ they imply $w = 0$; for $a = 1$ they imply $w = G$. $\square$

For $A \subseteq \mathcal{I}$, let $R_A = \sum_{i \in A} r_i$. Triangle inequalities for pairwise absolute deviations yield

$$|nR_A - |A|S| \le \sum_{i \in A, j \notin A} h_{ij} \le H \le n\gamma^U S. \tag{11}$$

Thus the subset-envelope and local centering rows used by the static model are valid. In particular, setting $A = \{i\}$ gives $|nr_i - S| \le \sum_{j \ne i} h_{ij}$. The low-Gini $L_1$ auxiliaries implement the same absolute-deviation logic without changing the feasible integer set.

Let $r_{\min} \le r_i \le r_{\max}$ for all $i$. Since $H \ge (n-1)(r_{\max} - r_{\min})$, the active variable-$S$ centering row

$$(n-1)(r_{\max} - r_{\min}) \le n\gamma^U S \tag{12}$$

follows from (8).

### 3.2 Objective estimators

Let $P^L \le P \le P^U$ and $S^L \le S \le S^U$ be valid model-domain bounds. The active $W_{SP}$ variable satisfies the four McCormick rows for $SP$ on this box. Every original point admits $W_{SP} = SP$ and hence satisfies

$$H + n\lambda W_{SP} \le n(\overline{F} - \varepsilon)S. \tag{13}$$

Consequently, the McCormick system plus (13) is a valid relaxation of the nonlinear cutoff. The simpler active estimator

$$H + nS^U \lambda P \le nS^U(\overline{F} - \varepsilon) \tag{14}$$

is necessary because $H/(nS) + \lambda P \le \overline{F} - \varepsilon$, $S \le S^U$, and $P \ge 0$. Domain minima of $|Y_i/t_i - 1|$ give $P \ge P^L$; if $\gamma^L + \lambda P^L \ge \overline{F} - \varepsilon$, the leaf is empty of improvers.

### 3.3 Inventory, movement, and visit rows

Vehicles start empty and may unload residual pickup at the depot. Therefore

$$\sum_i b_i - \sum_k Q_k \le \sum_i Y_i \le \sum_i b_i. \tag{15}$$

For station $i$ and vehicle $k$, define the unavoidable return travel $\rho_{ki} = \tau_{0i} + \tau_{i0}$ and $m_{ki} = \lfloor (T_k - \rho_{ki})/c^{\text{unit}} \rfloor_+$. Station disjointness implies the safe reachability domain

$$Y_i \ge b_i - \max_k \min\{b_i, Q_k, m_{ki}\}, \tag{16}$$

$$Y_i \le b_i + \max_k \min\{c_i - b_i, Q_k, m_{ki}\}. \tag{17}$$

Intersecting these bounds with capacities and the incumbent penalty budget is valid domain propagation. The visit links

$$Y_i - b_i \le (c_i - b_i) \sum_k z_{ki}, \tag{18}$$

$$b_i - Y_i \le b_i \sum_k z_{ki} \tag{19}$$

force $Y_i = b_i$ when no vehicle visits. Required-movement rows merely sum the minimum pickup or drop implied by these valid final-inventory domains, so they are necessary. The global handling row $c^{\text{unit}} \sum_{k,i} p_{ki} \le \sum_k T_k$ follows by summing (2) and dropping nonnegative travel.

For any pair $i, j$, valid ratio domains directly bound $h_{ij} = |r_i - r_j|$; these are the active Gini spread rows. Subset inventory rows similarly bound the net pickup or delivery of a station subset by the sum of valid station and vehicle movement capacities. They cannot exclude an original route-load plan because every such plan obeys the same conservation.

### 3.4 Duration and transfer compatibility

For a small support $A$ and vehicle $k$, let $L_k(A)$ be a lower bound on the depot tour visiting every station in $A$. If

$$L_k(A) + c^{\text{unit}}\lceil |A|/2 \rceil > T_k, \tag{20}$$

then $\sum_{i \in A} z_{ki} \leq |A| - 1$ is valid. The tested model generates this family for pairs and triples, using exact small-subset cycle bounds and a conservative handling lower bound.

Under the empty-start convention, positive net delivery by vehicle $k$ into $D \subseteq \mathcal{I}$ must be sourced by pickups outside $D$:

$$\sum_{j \in D} d_{kj} - \sum_{j \in D} p_{kj} \leq \sum_{i \notin D} p_{ki}. \tag{21}$$

The pairwise compatibility rows further remove a source–receiver pairing only when a conservative depot–source–receiver–depot duration lower bound proves that pairing impossible. Both statements are necessary for every original route of an empty-start vehicle.

Receiver-set cover, diagnostic lower/equality rows for $H$, GS coupling, and disaggregated SP coupling are not active in the tested profile and supply no paper evidence here.

## 4 CPLEX-Managed Tailored Branch-and-Cut

The fixed-interval compact model is solved through the CPLEX C API [2]. CPLEX manages presolve, LP relaxations, branching, node selection, incumbents, best bounds, and MIP termination. The tailored layer constructs the original fixed-interval model, applies the static valid rows of the previous section, and registers a bounded user-cut separator.

### 4.1 Selected directed route-cutset separator

For vehicle $k$, a station subset $A \subseteq \mathcal{I}$ not containing the depot, and a representative $\ell \in A$, define the directed boundary

$$\delta_k(A) = \sum_{i \notin A, j \in A} x_{kij} + \sum_{i \in A, j \notin A} x_{kij}. \tag{22}$$

The callback separates

$$\delta_k(A) \geq 2 z_{k\ell}. \tag{23}$$

**Proposition 7** (Validity of the callback route cut)**.** *Inequality* (23) *is valid for every original integer route, regardless of how $A$ was selected.*

*Proof.* If $z_{k\ell} = 0$, the right-hand side is zero and the inequality follows from nonnegativity. If $z_{k\ell} = 1$, the depot-rooted route visits a node in $A$. Because the depot is outside $A$ and the route starts and ends at the depot, it must enter $A$ at least once and leave $A$ at least once. These two arcs contribute at least two to $\delta_k(A)$. □

At a relaxation callback, stations with positive fractional visit values are ranked. Candidate subsets have size at most four, at most 50 cuts are added per fixed-interval solve, and only rows violated by more than $10^{-6}$ are submitted. The relaxation vector is therefore a separation heuristic only; the proof above, not the sampled vector, establishes validity. No candidate selection rule reads an instance name, seed, path, archived result, or known objective.

The implementation records candidate count, violated count, added count, maximum and mean violation, and cut counts by subset size. It also exports the first relaxation vector for diagnostic fractionality analysis. This vector never enters the certificate ledger.

### 4.2 Solver boundary and exact closure

CPLEX best bounds are admissible only for the original fixed-interval compact model with paper-safe rows. Integer infeasibility closes a no-improver leaf; integer optimality plus independent route reconstruction may update a same-run upper bound according to the sealed-run policy. A finite time-limit best bound may improve the ledger only when its scope and validity flags identify the original fixed interval. Wrapper-only heartbeats and diagnostic vectors cannot close a leaf.

The official plain benchmark uses the current binary-expansion compact MILP and CPLEX with the same single-thread policy. It does not use the frontier, tailored static rows, or route callback and is never imported into the tailored certificate. Standard branch-and-cut validity follows the requirement that user cuts retain every original integer feasible point [3].

## 5 Certificate Ledger and Evidence Boundary

Each final Gini leaf records its interval, lower bound, closure basis, scope, solver status, and whether exact coverage remains. A global certificate is written only when the verified incumbent exists, the improving domain is covered, every final leaf is validly closed, no invalid interval or open node remains, and the aggregate lower bound reaches the incumbent within tolerance.

The admissible sources are valid frontier/relaxation bounds, exact original fixed-interval infeasibility, valid original fixed-interval CPLEX best bounds, and empty-domain proofs. Same-run heuristic routes are upper bounds only and must pass independent route, load, inventory, Gini, penalty, and objective verification.

The tested commands disable archive scanning, imported known or external incumbents, focus-only closure, route-subset enumeration, manual fixed-interval upper-bound import, and the optional S ledger. Plain CPLEX rows, callback vectors, telemetry, and any diagnostic model are benchmark or diagnostic artifacts only. The full-frontier merge does not read them.

Audits enforce these rules at three levels. Source audits reject instance-specific algorithm branches. Per-JSON certificate audits check scope, coverage, verifier status, and forbidden-source flags. Package audits require fresh-run provenance and reject summary or raw-result pointers to a different result round. Thread and objective-convention audits ensure that tailored and benchmark rows use one CPLEX thread and the same $\lambda$, duration, inventory, target, Gini, and penalty conventions.

**Corollary 8** (No diagnostic contamination)**.** *If the certificate-source, package-locality, and paper-strict audits pass, and the ledger hypotheses of the full-frontier theorem hold, no benchmark or diagnostic row is needed to establish the reported optimum.*

*Proof.* The allowed closure-source enumeration is exhaustive in the ledger audit. Every disallowed source is either absent or explicitly marked noncontributing; the audit rejects any contrary row. The remaining accepted sources are those used in the theorem. □

# 6 Computational Protocol

The fresh evaluation uses source commit recorded in the package manifest and stores every raw JSON, child interval model, progress trace, solver log, and summary under one result directory. No prior raw row or fixed-interval incumbent is copied. Tailored and official plain CPLEX rows both use one solver thread. A generic wall-budget policy allocates 68% to frontier construction, 28% to exact interval leaves, and 4% to model and wrapper overhead; the benchmark receives the matched nominal wall budget directly. Runs longer than three hours emit five-minute progress and process-tree memory samples.

The controls are V12/M1, V12/M2, `tight_T_seed3101`, and `high_imbalance_seed3202`. The hard set is `moderate_seed3301`, `moderate_seed3302`, `high_imbalance_seed3201`, and `tight_T_seed3102`. Comparisons use matched nominal budgets and distinguish certified runtime from noncertified bound/gap evidence. Fixed-interval diagnostics are not compared as if they were full-frontier runs.

## 6.1 Fresh selected results

| Instance | Method | Budget | LB | UB | Gap | Status |
|---|---|---|---|---|---|---|
| V12_M1 | Tailored | 1200 | 0.357201 | 0.357201 | 0 | optimal |
| V12_M1 | Plain | 300 | 0.357201 | 0.357201 | 0 | optimal |
| V12_M2 | Tailored | 1200 | 0.718504 | 0.718504 | 0 | optimal |
| V12_M2 | Plain | 1200 | 0.651265 | 0.718504 | 0.09358 | not_certified |
| tight_T_seed3101 | Tailored | 300 | 0.107253 | 0.107253 | 0 | optimal |
| tight_T_seed3101 | Plain | 1200 | 0.0191812 | 0.134542 | 0.8574 | not_certified |
| high_imbalance_seed3202 | Tailored | 1200 | 1.74931 | 1.74931 | 0 | optimal |
| high_imbalance_seed3202 | Plain | 1200 | 1.1723 | 1.78996 | 0.3451 | not_certified |
| moderate_seed3301 | Tailored | 14400 | 0.0122881 | 0.0491526 | 0.75 | gcap_frontier_not_closed |
| moderate_seed3301 | Plain | 14400 | 0.0416462 | 0.049369 | 0.1564 | not_certified |
| moderate_seed3302 | Tailored | 14400 | 0.0366818 | 0.195636 | 0.8125 | gcap_frontier_not_closed |
| moderate_seed3302 | Plain | 3600 | 0.121508 | 0.230179 | 0.4721 | not_certified |
| high_imbalance_seed3201 | Tailored | 14400 | 1.74211 | 2.4434 | 0.287 | gcap_frontier_not_closed |
| high_imbalance_seed3201 | Plain | 14400 | 2.12338 | 2.44369 | 0.1311 | not_certified |
| tight_T_seed3102 | Tailored | 14400 | 0.450176 | 0.600704 | 0.2506 | wrapper_error_noncertified |
| tight_T_seed3102 | Plain | 14400 | 0.49525 | 0.618593 | 0.1994 | not_certified |

Rows are selected from this round only. Certified rows use the smallest fresh budget that certified; otherwise the longest valid row is shown. Plain rows are benchmark-only.

In this matrix the tailored framework certifies all four controls, although the required budget is not uniformly lower than plain CPLEX. None of the four hard rows is certified. The long hard rows expose both valid open-ledger progress and wrapper/native-exit failures; these are reported rather than converted into closures. Size-four callback rows are observed, confirming that the selected separator is active, but this experiment has no matched no-callback tailored arm. It therefore does not identify the callback's causal contribution separately from the rest of the tailored framework. The mixed hard-row comparison supports continued experimental use of the profile, not a claim of uniform dominance or a universal default policy.

The computational conclusions are deliberately limited to this matrix. A certified control supports stability; an open hard row reports its valid ledger bound and blocker. A route-profile advantage requires either an exact certificate absent from the matched benchmark or a clearly stronger valid same-budget bound trajectory. No claim is inferred from callback row count alone.

## References


[1] Corrado Gini. Variabilita e mutabilita. *Studi Economico-Giuridici della R. Universita di Cagliari*, 1912.

[2] IBM. *IBM ILOG CPLEX Optimization Studio 22.1.1 Documentation*, 2022.

[3] Laurence A. Wolsey. *Integer Programming*. Wiley, 1998.